ABESIT

Project Id:(……)

Min Project Synopsis

on

Title Name

Submitted in Partial Fulfillment of the Requirement

For the Degree of

Bachelor of Technology

In

Computer Science and Engineering

By

Name of the Student 1 (Roll Number 1)
Name of the Student 2 (Roll Number 2)
Name of the Student 3 (Roll Number 3)
Name of the Student 3 (Roll Number 3)

Under the Supervision

of

……

Designation
Department of Computer Science Engineering

ABES INSTITUTE OF TECHNOLOGY, GHAZIABAD

AFFILIATED TO

Dr.A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW

(ODD SEM, 2026-27)

Table of Contents

Chapter No.	Contents	Page No.
1	Abstract	3
2	Introduction	4
3	Literature Survey	5-7
4	Problem Statement	8
5	Proposed Methodology	9-10
6	Implementation & Results	11-12
7	Mapping with SDG Goal	13
8	Conclusion/Future Scope	14
9	References	15-16

Abstract

(at least 150-200 words)

- What is the problem / purpose?
- How did you do the work (methods)?
- What did you find (results)?
- What do the results mean (conclusion/impact)?

Keywords:

.

Introduction

(At least 1-2 Pages)

Literature Survey

(Literature survey should be prepare from 5 latest papers)

S. No.	Author(s) & Year	Title of Paper / Work	Objective / Focus	Methodology / Approach	Key Findings
1	Author et.al. (2025)	Study on IoT-based Smart Home	To develop a cost-effective IoT-based automation system	Used sensors (temperature, motion), Arduino, Wi-Fi module	Improved energy efficiency and security
2	Author et.al (2024)	Machine Learning for Disease Prediction	Apply ML models for early disease detection	Implemented Decision Tree, SVM, and ANN with medical dataset	ANN achieved highest accuracy (92%)
3	Author et.al. (2023)	Renewable Energy Optimization	Optimize solar panel energy output	MATLAB simulations with varying tilt angles	Achieved 15% efficiency improvement
4	Author et.al.(2022)	Cloud Computing Security Issues	Identify vulnerabilities in cloud services	Literature review of existing studies	Highlighted major issues: data leakage, insider threats

Problem Statement

- Description of the Problem Statement
- Objectives

Proposed Methodology

(Tech Stack, Block Diagrams and Flow charts)

Implementation & Results

SDG Goals Mapping

Conclusion and Future Scope

References (In APA Format)

[1] Ritu, Raj, Sondhiya. (2024). Smart Solutions for Sustainable Agriculture: Design and Development of IoT-Powered Soil Health Identification Systems. Deleted Journal, 31(4s):510-519. doi: 10.52783/cana.v31.946

[2] S., R., Jino, Ramson., Walter, D., Leon-Salas., Zachary, Brecheisen., Erika, J., Foster., Cliff, T., Johnston., Darrell, G., Schulze., Timothy, R., Filley., Rahim, Rahimi., Martin, Villalta, Soto., J., A., L., Bolivar., Mauricio, Postigo, Malaga. (2021). A Self-Powered, Real-Time, LoRaWAN IoT-Based Soil Health Monitoring System. IEEE Internet of Things Journal, 8(11):9278-9293. doi: 10.1109/JIOT.2021.3056586

[3] Jordi, Surrallés. (2022). ETS: A Smart and Enhanced Topsoil Health Monitoring and Control System at Edge using IoT. doi: 10.1109/ises54909.2022.00153

[4] Laavanya, Rachakonda. (2022). ETS: A Smart and Enhanced Topsoil Health Monitoring and Control System at Edge using IoT. 689-693. doi: 10.1109/iSES54909.2022.00153

[5] Smieee, C.Rajasekaran., Ece, K.S.Rangasamy, Professor., Ece, M.Jeevanantham, Student., Ece, R.Jeeva, Student., Smieee, K.B.Jayanthi., Ece, R.Hariharan, Student., K.Raguvaran. (2024). An Integrating IoT for Intelligent Soil Analysis and Crop Optimization. 1-4. doi: 10.1109/aimla59606.2024.10531447

[6] Huynh,, Ethan., Pablo, Ramírez., Olimboyeva, Malohat. (2022). Smart agricultural remote monitoring system for better soil health using IoT. International Journal of Health Sciences (IJHS), 1239-1251. doi: 10.53730/ijhs.v6ns8.9885

[7] R., Baskar., Golden, Kumar., Dragica, Karan. (2022). Smart agricultural remote monitoring system for better soil health using IoT. International Journal of Health Sciences (IJHS), doi: 10.53730/ijhs.v6ns4.9885

[8] Rajan, Prasad., Rajinder, Tiwari., Adesh, Kumar, Srivastava. (2023). Internet of Things-Based Fuzzy Logic Controller for Smart Soil Health Monitoring: A Case Study of Semi-Arid Regions of India. doi: 10.3390/ecsa-10-16181

[9] Vaibhav, Bhatnagar., Ramesh, Chandra. (2020). IoT-Based Soil Health Monitoring and Recommendation System. 1-21. doi: 10.1007/978-981-15-0663-5_1

[10] Sashant, Suhag., Nidhi, singh., Sanskriti, Jadaun., Prashant, Johri., Ayush, Shukla., Nidhi, Parashar. (2021). IoT based Soil Nutrition and Plant Disease Detection System for Smart Agriculture. 478-483. doi: 10.1109/CSNT51715.2021.9509719

[11] Holzinger, A., Keiblinger, K., Holub, P., &Zatloukal, K. (2023). AI for life: Trends in artificial intelligence for biotechnology. *New Biotechnology*. Elsevier. https://doi.org/10.1016/j.nbt.2023.01.001

[12] Giannakidou, S., Radoglou-Grammatikis, P., Lagkas, T., &Sarigiannidis, P. (2024). Leveraging the power of IoT and AI in forest fire prevention, detection, and restoration: A comprehensive survey. *Internet of Things*, Elsevier. https://doi.org/10.1016/j.iot.2024.100523

[13] Selvanarayanan, R., Rajendran, S., &Algburi, S. (2024). Empowering coffee farming using counterfactual recommendation-based RNN-driven IoT integrated soil quality command system. *Scientific Reports*. https://doi.org/10.1038/s41598-024-56954-x

[14] Wang, R., Sun, Y., Zong, J., Wang, Y., Cao, X., & Wang, Y. (2024). Remote sensing application in ecological restoration monitoring: A systematic review. *Remote Sensing*. https://doi.org/10.3390/rs16122204

[15] Singh, S., & Jain, P. (2022). Applications of artificial intelligence for the development of sustainable agriculture. *Agro-biodiversity and Agri-ecosystem Management*. Springer. https://doi.org/10.1007/978-981-19-0928-3_16

[16] Robinson, J. M., &Annells, A. (2024). Monitoring soil fauna with ecoacoustics. *Proceedings of the Royal Society B: Biological Sciences*. https://doi.org/10.1098/rspb.2024.1595

[17] Sharma, C., Pathak, P., Kumar, A., & Gautam, S. (2024). Sustainable regenerative agriculture allied with digital agri-technologies and future perspectives for transforming Indian agriculture. *Environment, Development and Sustainability*. Springer. https://doi.org/10.1007/s10668-024-05231-y

[18] Akhtar, M. N., Shaikh, A. J., Khan, A., Awais, H., & Bakar, E. A. (2021). Smart sensing with edge computing in precision agriculture for soil assessment and heavy metal monitoring: A review. *Agriculture*. https://doi.org/10.3390/agriculture11060475

[19] Elbasi, E., Mostafa, N., AlArnaout, Z., &Zreikat, A. I. (2022). Artificial intelligence technology in the agricultural sector: A systematic literature review. *IEEE Access*. https://doi.org/10.1109/ACCESS.2022.9999442

[20] Ahmed, G. N., &Kamalakkannan, S. (2022). Developing an IoT-based data analytics system for predicting soil nutrient degradation level. *Expert Clouds and Applications*. Springer. https://doi.org/10.1007/978-981-16-2126-0_12